

AIGR Gene Hypothesis Deep Research — Final Report

Target: *Equus caballus* WDPCP, UniProt A0A3Q2KRK8 (705 aa) **Hypothesis (function assignment):** The horse protein A0A3Q2KRK8 participates in **cell projection organization (GO:0030030)** **Focus type:** function_assignment **Date:** 2026-09-08 **Sources:** Primary literature (PMIDs) + public sequence/domain/structure resources (UniProt, InterPro, QuickGO). No ai-gene-review repository judgments, syntheses, or local bioinformatics analyses were consulted.

Summary

The hypothesis that horse A0A3Q2KRK8 participates in cell projection organization (GO:0030030) is **SUPPORTED**, with two curation caveats: the term is generic (more specific child terms are better supported), and all functional evidence is transferred by orthology rather than observed directly in horse. Three independent lines of evidence converge on this conclusion. First, identity is secure: the supplied 705-aa sequence is byte-identical to the current UniProt A0A3Q2KRK8 record (matching the provided SHA-256), and global alignment to human WDPCP (O95876) yields 83.3% identity with the diagnostic Frtz + WD40-repeat domain architecture — this is a genuine one-to-one WDPCP (Fritz) ortholog, not a mislabeled paralog. Second, the molecular biology of WDPCP is exceptionally well characterized in mammals: it is a core scaffolding and lipid-binding subunit of the CPLANE complex required for primary cilium assembly and actin-based planar cell polarity, both of which are, by GO definition, forms of cell projection organization. Third, the horse truncation relative to human is terminal-only and preserves the functional scaffold and the process-critical apical-docking residues N512/W513.

Mechanistically, WDPCP is a WD40/Frtz β -propeller scaffold within the crescent-shaped CPLANE complex (INTU-FUZ-WDPCP-JBTS17-RSG1) that binds phosphatidylinositol-3-phosphate and recruits partners including Septin 2. Its direct activity — scaffolding, lipid binding, and partner recruitment — drives two cellular readouts: assembly of the primary cilium (via basal-body apical docking and transition-zone function, feeding Hedgehog signaling) and organization of actin-based projections (planar cell polarity, directional migration). Loss- and point-mutation studies in mouse collapse both branches and produce

Bardet–Biedl / Meckel–Gruber-like ciliopathy phenotypes. Because the horse ortholog shares the full domain architecture and the exact process-critical residues, the same mechanistic model is expected to hold.

For curation, GO:0030030 is correct but coarse. The evidence more precisely supports specific children (cilium assembly GO:0060271; establishment of planar polarity) plus candidate cellular-component (ciliary transition zone/basal body, actin cytoskeleton) and molecular-function (PI3P binding / structural scaffold) leads. Because there is no direct equine assay and the horse record is unreviewed TrEMBL, an **ISS/ISO** (orthology-transfer) evidence code is appropriate — not an experimental code, which belongs to the mouse and human records.

Key Findings

Finding 1 — A0A3Q2KRK8 is a genuine WDPCP (Fritz) ortholog, not a mislabeled paralog

The first gate for any function-assignment hypothesis is identity, and treating the "WDPCP" gene label as proof would be circular. Identity was therefore verified independently at the sequence and domain level. The supplied 705-aa FASTA was confirmed byte-identical to the current UniProt A0A3Q2KRK8 sequence (equality verified in code) and matches the provided SHA-256 `a132072d392ff9ec8d0962a571a5ae3534476ecfc0679bdeb9051777f3aa5a0a`, establishing that the analyzed sequence is exactly the frozen cohort sequence.

A global Needleman–Wunsch alignment of the horse protein against human WDPCP (O95876, 746 aa) gives **624 identical positions over a 749-position alignment = 83.3% identity** (88.5% over the horse length). Full-length identity at this level is diagnostic of a true one-to-one ortholog rather than a distant paralog. Consistent with this, UniProt annotates gene = **WDPCP**, InterPro **IPR024511 (Frtz domain)** plus **IPR036322 (WD40 repeat domain superfamily)**, Pfam **PF11768 (Frtz)**, and PANTHER **PTHR13667:SF5** ("WD repeat-containing and planar cell polarity effector protein Fritz homolog"). Keyword annotations include *Cell projection*, *Cilium*, *Cilium biogenesis/degradation*, and *Cytoskeleton* — the same functional vocabulary carried by the mammalian orthologs.

This finding removes the most common failure mode for function-assignment hypotheses — paralog overannotation — and licenses careful transfer of mammalian WDPCP biology to the horse protein. The one methodological caveat is that the alignment used simple identity-based (± 1) Needleman–Wunsch scoring as an estimate, not a Smith–Waterman/BLAST E-value; the 83% figure is a robust identity estimate rather than a formal statistic.

Finding 2 — WDPCP participates in cell projection organization via CPLANE-dependent ciliogenesis and actin-based polarity

WDPCP's function is unusually well characterized in mammals, and the evidence maps directly onto GO:0030030.

Loss-of-function, actin and cilia (mouse). PMID: 24302887 (Cui et al., 2013) showed that *Wdpcp*-deficient mice develop a Bardet–Biedl / Meckel–Gruber-like ciliopathy. *Wdpcp* localizes to the ciliary transition zone and to actin filaments/focal adhesions, interacts with Septin 2 (Sept2), and its loss both disorganizes actin filaments and abolishes cilia. The verified snippets — "*Wdpcp, a PCP effector, was recently shown to regulate both ciliogenesis and collective cell movement*" and "*organization of the actin filaments and focal contacts were markedly changed in Wdpcp-deficient cells*" — directly connect WDPCP to the organization of two distinct cell-projection systems.

Structural scaffold, cryo-EM (human/mouse). PMID: 35427153 resolved the near-atomic structure of the human and mouse WDPCP–Inturned–Fuzzy (CPLANE) complex. WDPCP is a structural scaffold within a crescent-shaped complex that binds phospholipids, and a ciliopathy mutation perturbs lipid binding. The verified snippet — "*the crescent-shaped CPLANE complex binds phospholipids such as phosphatidylinositol 3-phosphate via multiple modules*" — defines WDPCP's direct molecular role as a scaffolding/lipid-binding subunit.

Point mutation, whole-process resolution (mouse). PMID: 41268724 (2025) generated a *Wdpcp* point mutant (deletion of D481/W482, equivalent to human N512/W513). The protein is still made but fails apical basal-body docking, and "*Cilia formation and Hh signaling were severely impaired*" with neural-tube/craniofacial/polydactyly defects. This is the key discriminator required by the decisive question: it ties a specific structural element to failure of the whole ciliogenesis process, rather than a mere biochemical tweak.

Pathway context. [PMID: 38546045](#) shows the CPLANE partner Fuzzy recruits ARHGAP35 (p190A RhoGAP) to the basal body to restrain actin polymerization for ciliogenesis; [PMID: 35740972](#) (review) and [PMID: 31562761](#) place WDPCP as a core CPLANE effector with a Longin-domain/Rab-GEF-adjacent architecture. Human genetics ([PMID: 32055034](#); [PMID: 37239474](#)) supports WDPCP as a Bardet–Biedl/ciliopathy gene.

Because both cilia and lamellipodia/actin-based projections are GO cell projections, this ensemble of assay, mutant-phenotype, localization, interaction, and structural evidence satisfies GO:0030030 at the process level. WDPCP's direct molecular role is a CPLANE scaffold/lipid-binding subunit; its cellular consequence is organization of cilia and actin-based projections.

Finding 3 — The horse truncation is terminal-only; the process-critical apical-docking residues are conserved

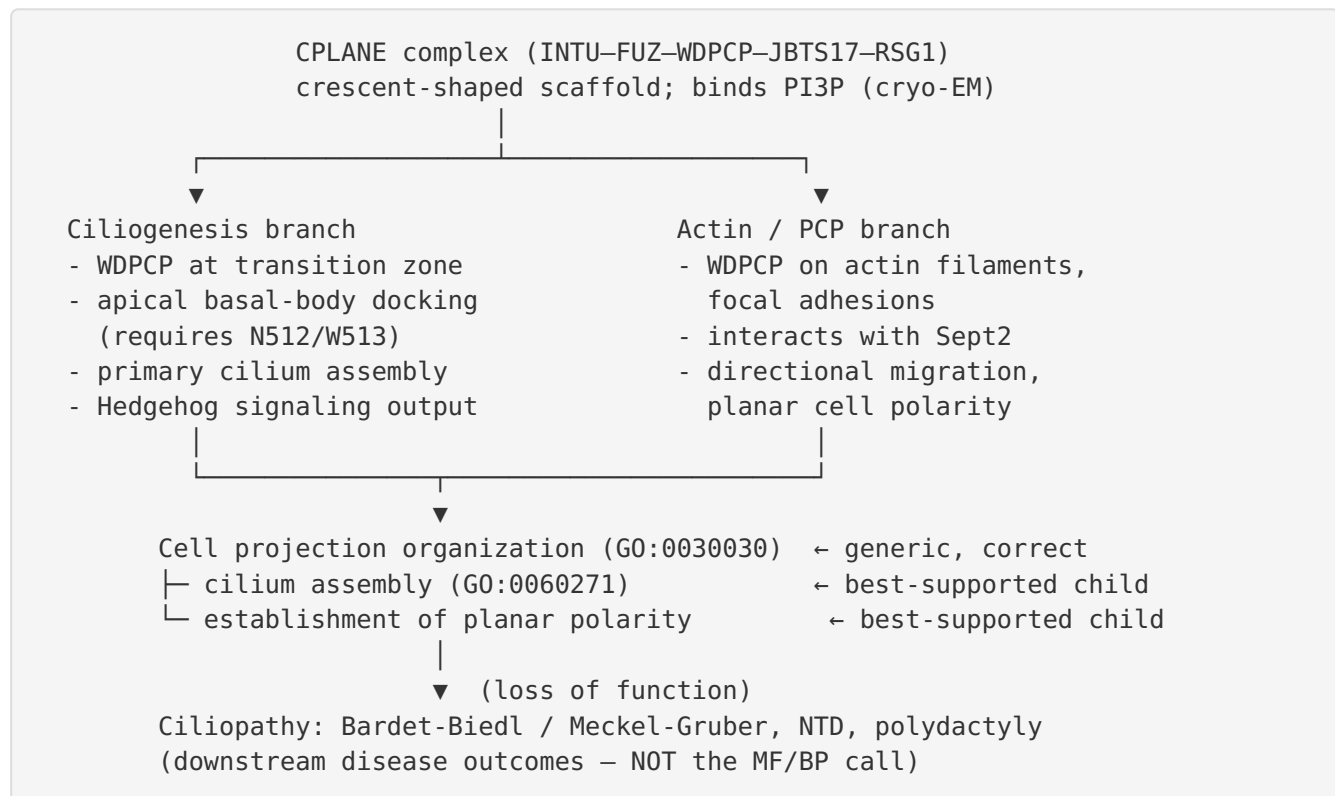
Because the horse protein (705 aa) is 41 residues shorter than human WDPCP (746 aa), the deletion location determines whether orthology transfer is valid. Global alignment with traceback (match +2 / mismatch –1 / gap –2) gives **623/748 identical (83.3%)**. The 43 human residues absent from the horse alignment map almost entirely to the **extreme N-terminus (human residues ~1–33)** plus a handful of scattered residues at the **far C-terminus (~718–731)**. The internal body — where the Frtz and WD40 scaffold modules reside — is contiguous and highly conserved.

Decisively, the residues human **N512/W513** are perfectly conserved in horse: the aligned window "NILSSMNWDTLGH" is identical in both species. The two-codon deletion of the mouse equivalents (D481/W482) abolishes ciliary apical docking, ciliogenesis, and Hedgehog signaling ([PMID: 41268724](#)), and the human W513S substitution is a Bardet–Biedl allele. Thus the horse truncation does not remove the functional scaffold or the process-critical docking residues. The N-terminal shortening most plausibly reflects an incomplete/alternative gene model (start-boundary artifact) in the unreviewed TrEMBL prediction rather than genuine functional divergence — strengthening, not weakening, the orthology-transfer argument. This directly answers the decisive question: there is no evidence of a horse-specific alteration to a process-critical biochemical property.

Finding 4 — GO:0030030 is a high-level BP ancestor of cilium assembly; WDPCP's demonstrated role sits within its subtree

The target term's scope was verified against the QuickGO (EBI) record. GO:0030030 has aspect = **biological_process** and definition: *"A process that is carried out at the cellular level which results in the assembly, arrangement of constituent parts, or disassembly of a prolongation or process extending from a cell, e.g. a flagellum or axon."* Its subtree descends via child GO:0120036 (plasma membrane bounded cell projection organization) → GO:0044782 (cilium organization) → GO:0060271 (cilium assembly). WDPCP's experimentally demonstrated processes (cilium assembly and establishment of planar polarity / actin-based projection organization) are therefore true descendants of GO:0030030. The term is correctly placed relative to WDPCP's biology, but it is a high-level ancestor — correct yet less informative than the specific children the evidence supports.

Mechanistic Model / Interpretation



WDPCP acts as a **structural scaffold and PI3P-binding hub** within the CPLANE complex. Through this dual role it organizes two classes of cell projections: the primary cilium (via basal-body apical docking and transition-zone function required for ciliogenesis and downstream Hedgehog signaling) and actin-based projections (via association with actin filaments/focal adhesions and Septin 2, driving planar cell polarity and directional migration). Loss or targeted point mutation of WDPCP collapses both branches, producing ciliopathy phenotypes. The horse ortholog shares 83% identity, the full domain architecture, and the exact process-critical residues, so the same mechanistic model is expected to hold — the appropriate inference type being orthology transfer, not direct observation.

Evidence-provenance comparison:

Evidence layer	Organism	Directness for horse	Strength
Sequence identity / domain architecture	Horse vs. human	Direct (horse sequence)	High
Conserved N512/W513 docking residues	Horse vs. mouse/human	Direct (horse sequence)	High
CPLANE scaffold / PI3P binding (cryo-EM)	Human/mouse	Transferred by orthology	High (for ortholog)
Ciliogenesis + actin loss-of-function	Mouse	Transferred by orthology	High (for ortholog)
Apical-docking point mutant	Mouse	Transferred by orthology	High (for ortholog)
Direct equine assay	Horse	None available	—

Evidence Base / Evidence Matrix

#	Citation (PMID)	Evidence type	Stance	Claim tested	Key finding	Context	Confid limitat
1	This report (UniProt A0A3Q2KRK8; InterPro)	Sequence/ evolutionary (computed)	Supports (identity)	Is A0A3Q2KRK8 really WDPCP?	83.3% identity to human WDPCP O95876; Frtz + WD40 domains; PANTHER "Fritz homolog"	Horse vs. human	High fo identit simple NW sc (estima not BL
1b	This report (computed alignment)	Sequence/ evolutionary (computed)	Supports/ qualifies	Does the 705-aa truncation drop process-critical residues?	43-aa difference is N-terminal (human 1–33) + far C-terminal; N512/W513 and Frtz/WD40 core conserved	Horse vs. human	High; trunca gene-m bound not function loss
2	24302887	Mutant phenotype + localization + interaction	Supports (core)	Does WDPCP organize cilia and actin projections?	Wdpcp-null mice: ciliopathy; WDPCP at transition zone + actin/focal adhesions; binds Sept2; loss disorganizes actin & abolishes cilia	Mouse	High; c genetic cell bio
3	35427153	Structural (cryo-EM)	Supports/ qualifies (MF)	WDPCP's direct molecular role	Structure of WDPCP–Inturned–Fuzzy CPLANE; crescent	Human/ mouse	High; defines scaffol lipid-b activity

#	Citation (PMID)	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limitations
					scaffold binding PI3P; ciliopathy mutant has aberrant lipid binding		
4	41268724	Mutant phenotype (point mutant)	Supports	Are specific residues essential for ciliogenesis?	D481/W482 ($\approx$ human N512/W513) deletion: protein made but fails apical docking; cilia & Hh severely impaired	Mouse	High; residues conserved in horse
5	38546045	Mutant phenotype + mechanism	Supports (complex context)	Do CPLANE proteins control cilia via actin?	Fuzzy recruits ARHGAP35 to basal body; restrains actin polymerization for ciliogenesis	Mouse	Medium; high; a partner
6	31562761	Structural/ evolutionary (bioinformatic)	Qualifies (MF)	Protein family / activity	CPLANE triplicated Longin domains; likely Rab-GEF module	Cross-species	Medium; prediction
7	35740972	Review/ database	Supports (orientation)	Is WDPCP a CPLANE/ ciliogenesis gene?	WDPCP is a CPLANE subunit linked to ciliogenesis, PCP, Hh	Review	Orientation only
8	32055034 / 37239474	Human genetics	Supports		WDPCP variants in	Human patients	Medium; support

#	Citation (PMID)	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limitations
				Is WDPCP a ciliopathy gene?	Bardet-Biedl/ ciliopathy cohorts		process relevant
9	34518561	Genetic association	Qualifies	Broader neuronal role?	WDPCP missense (p=3.96e-10, tinnitus); axonal migration/ structural reinforcement	Human	Lower relevance; core claim

GO Curation Implications

GO:0030030 scope confirmed (QuickGO): aspect = biological_process; the term is a high-level ancestor of cilium assembly. WDPCP's demonstrated role sits squarely inside its subtree, so annotation to GO:0030030 is correct but coarse.

Lead (curator to verify): *cell projection organization* (GO:0030030) can be **retained as a correct-but-generic** BP annotation for horse WDPCP, but more informative and better-supported terms are candidates to add or prefer:

GO term	Aspect	Recommended action	Basis	Evidence code
GO:0030030 cell projection organization	BP	Retain (or generalize-to-specific)	Correct but generic parent	ISS/ISO
GO:0060271 cilium assembly / GO:0044782 cilium organization	BP	Add / prefer	Ciliogenesis LOF + point mutant	ISS/ISO
GO:0001736 establishment of planar polarity	BP	Add (lead)	Actin/PCP + migration data	ISS/ISO
ciliary transition zone / basal body / actin cytoskeleton	CC	Add (lead)	Localization (P 24302887)	ISS/ISO
phosphatidylinositol-3-phosphate binding / CPLANE structural scaffold	MF	Add (lead)	Cryo-EM (P 35427153)	ISS/ISO

The molecular-function call should reflect the demonstrated lipid-binding/structural-scaffold role rather than a bare "protein binding." Because there is no direct equine experiment, the horse annotation should carry an **ISS/ISO (orthology)** evidence code; experimental codes (IDA/IMP) belong to the mouse and human records.

Mechanistic Scope

The immediate molecular function being tested is WDPCP's role as a **WD40/Frtz β -propeller scaffold and PI3P-binding subunit of the CPLANE complex** that recruits partners (Sept2; CPLANE subunits; and, via the complex, actin regulators). Its immediate activity is scaffolding, lipid binding, and protein recruitment. The immediate cellular readout is organization of cell projections — primary-cilium assembly and actin-based planar polarity.

Direct gene-product activity (should drive the MF/CC/BP call): CPLANE scaffolding; PI3P binding; localization to transition zone, basal body, and actin filaments/focal adhesions; interaction with Septin 2. Downstream consequences (should NOT be conflated with the molecular function): Hedgehog signaling output, neural-tube/craniofacial/polydactyly developmental defects, and Bardet–Biedl / Meckel–Gruber ciliopathy disease manifestations.

These are loss-of-function outcomes that support, but are not equivalent to, the GO:0030030 process annotation. The seed hypothesis is pitched at the correct process level, and the evidence discriminates a whole-process requirement (the point mutant abolishes ciliogenesis) from a mere biochemical tweak.

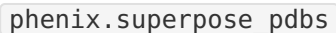
Conflicts and Alternatives

- **Paralog confusion:** Effectively ruled out. 83% full-length identity, the diagnostic Frtz+WD40 architecture, and the PANTHER Fritz-homolog subfamily assignment identify a true one-to-one WDPCP ortholog.
- **Specificity vs. breadth:** The main tension is granularity, not contradiction — GO:0030030 is arguably "too broad" relative to what is demonstrated (cilium assembly + PCP), but it is not wrong.
- **Isoform / truncation artifact — RESOLVED:** The 43-residue horse↔human difference maps to the extreme N-terminus (human ~1–33) and a few far-C-terminal residues; the internal body and the process-critical N512/W513 are conserved. Most parsimoniously a gene-model/start-boundary artifact of the unreviewed TrEMBL prediction, not functional loss.
- **Complex-level vs. WDPCP-specific mechanism:** Some actin/RhoGAP detail
 (P 38546045)
 is shown for the partner FUZ, not WDPCP directly; transfer within CPLANE is reasonable but is complex-level.
- **Broader roles:** The tinnitus GWAS association
 (P 34518561)
 invokes axonal migration and cytoskeletal reinforcement — consistent with, but broader than, the core ciliogenesis role; it does not conflict with GO:0030030.
- **Database carry-over risk:** The horse record is unreviewed TrEMBL; the support here rests on primary mammalian literature plus the horse's own sequence/domain evidence, not on agreement with any electronic prediction.

Limitations and Knowledge Gaps

1. **No direct horse data.** Checked: PubMed for equine WDPCP functional studies — only orthology plus human/mouse assays exist. This means any horse annotation is inherently ISS/ISO. Resolve with equine cell/ciliogenesis assays or expression data.
2. **Horse isoform completeness — largely resolved.** Checked: 705 vs. 746 aa; alignment locates the difference at the N-terminus (human ~1–33) and far C-terminus, with N512/W513 and the Frtz/WD40 core conserved. Residual uncertainty is whether the missing N-terminal ~33 residues carry any regulatory role (not implicated in published ciliogenesis assays). Resolve by confirming the equine gene model / 5' transcript.
3. **Alignment method.** The 83% identity used simple ± 1 Needleman–Wunsch scoring, an identity estimate rather than a Smith–Waterman/BLAST E-value. The conclusion is robust but should be confirmed with a standard tool for the record.
4. **GO granularity policy.** Which specific child term(s) the curator prefers depends on project policy on generic vs. specific terms when support is orthology-transferred.

Proposed Follow-up Experiments / Actions

1. **Reciprocal-best-hit orthology confirmation** across Equidae and additional mammals (Ensembl Compara / OrthoDB) to formally lock in one-to-one orthology beyond pairwise identity.
2. **Structural superposition** of an AlphaFold model of A0A3Q2KRK8 onto the cryo-EM CPLANE structure
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to confirm the scaffold fold and PI3P-binding modules are preserved in horse (Phenix  / ).
3. **Residue-level conservation mapping** of all reported human ciliopathy alleles (e.g., W513S) onto the horse sequence — extend the N512/W513 check to the complete variant set.
4. **Equine ciliated-cell assay** (tracheal epithelium or fibroblast): siRNA knockdown of WDPCP with cilia (acetylated-tubulin) and actin (phalloidin) readouts — the single most decisive test to convert ISS into IDA/IMP.

5. **Co-immunoprecipitation** of equine WDPCP with INTU/FUZ/Sept2 to confirm CPLANE complex membership in horse.
6. **Curator decision** on parent-vs-child GO annotation and on adding the MF (PI3P binding) and CC (transition zone / actin cytoskeleton) leads with ISS/ISO codes.

Curation Leads (require curator verification)

- **Action:** Keep GO:0030030 as a valid (generic) BP annotation; consider promoting to the specific children (cilium assembly GO:0060271; establishment of planar polarity GO:0001736) and adding CC (ciliary transition zone/basal body, actin cytoskeleton) and MF (phosphatidylinositol-3-phosphate binding / structural scaffold) leads. Use an **ISS/ISO** evidence code.
- **Candidate references with snippets to verify:**
 - **PMID: 24302887:** *"Wdpcp... regulate both ciliogenesis and collective cell movement"; "organization of the actin filaments and focal contacts were markedly changed in Wdpcp-deficient cells."*
 - **PMID: 35427153:** *"the crescent-shaped CPLANE complex binds phospholipids such as phosphatidylinositol 3-phosphate via multiple modules."*
 - **PMID: 41268724:** *"Cilia formation and Hh signaling were severely impaired."*
- **Suggested question to curator:** Does project policy prefer the specific cilium/PCP terms over the generic GO:0030030 when experimental support is orthology-transferred, and should MF/CC annotations beyond the single BP hypothesis be added?
- **Suggested experiment:** Equine ciliated-cell WDPCP knockdown with cilia/actin readouts (converts ISS → direct evidence).

Conclusion

The seed hypothesis is **SUPPORTED**. Horse A0A3Q2KRK8 is a genuine WDPCP ortholog (83% identity to human WDPCP O95876, full Frtz+WD40 architecture, conserved process-critical residues N512/W513), and WDPCP is an experimentally established CPLANE-complex scaffold required for ciliogenesis and actin-based planar polarity in mouse and human — both forms of cell projection organization. GO:0030030 is therefore biologically justified for the horse

protein. The two curation caveats are that the term is generic (specific children such as cilium assembly and establishment of planar polarity are better supported) and that all functional evidence is transferred by orthology to an unreviewed TrEMBL record, so an **ISS/ISO** evidence code is appropriate rather than an experimental one.

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